Name It, Draw It

V. 03

GRADED EXERCISE

NAME	Francine Hanna B. Alvarez
SCHOOL	Bicol University
YEAR AND PROGRAM	BS CHE 2B
SUBJECT	CHEM 13 — Organic Chemistry (Lecture)

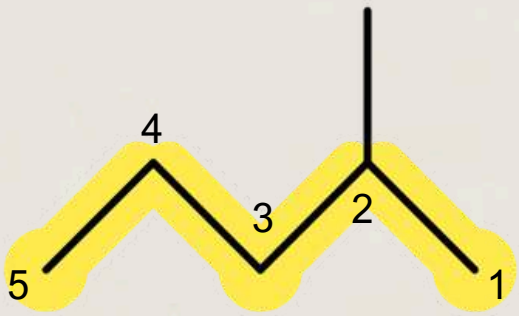
PART A

Structure → Name

AO1 – Foundation

Counted Formula: C_6H_{14}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 5 carbon atoms → pentane; open-chain structure.

Numbering: Left → right: 4; right → left: 2. Right → left is correct because 2 is the lower locant.

Substituents: Methyl at C2 → 2-methyl.

3 MY NAME (WRITTEN FIRST)

2-methylpentane

C_6H_{14}

4 THE ALKANE CHECK



2-methylpentane

Returned name: 2-methylpentane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

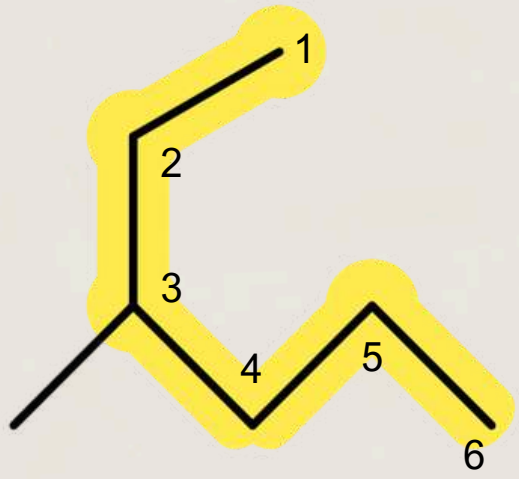
PART A

Structure → Name

A02 - Foundation

Counted Formula: C_7H_{16}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 6 carbon atoms → hexane. The chain bends at the branch point.

Numbering: Start from the ethyl end: methyl is at C3. Start from the other end: methyl is at C4.

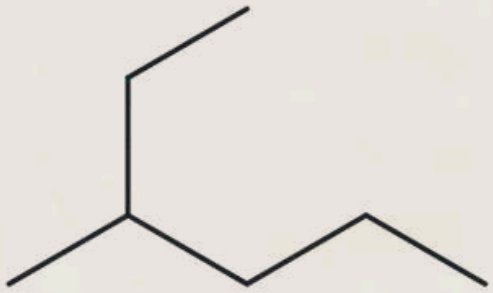
Substituents: Methyl at C3 → 3-methyl.

3 MY NAME (WRITTEN FIRST)

3-methylhexane

C_7H_{16}

4 THE ALKANE CHECK



3-methylhexane

Returned name: 3-methylhexane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

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PART A

Structure → Name

A03 - Foundation

Counted Formula: C_6H_{14}

PART A

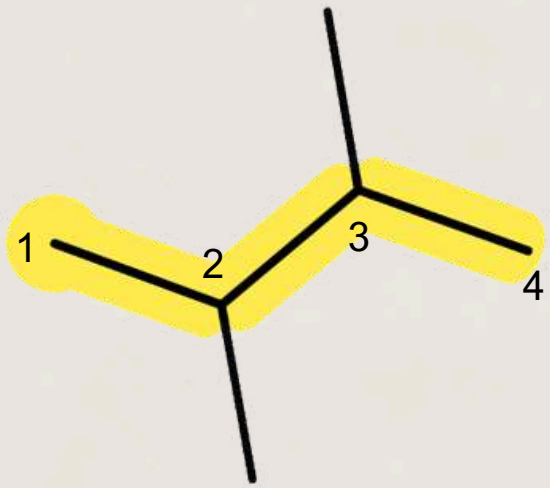
PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 4 C, so butane.

Numbering: both directions give 2,3, so it's a tie and either end works.

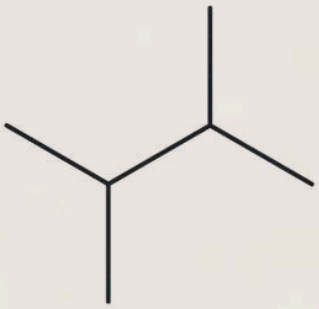
Substituents: methyl on C2 and C3, so 2,3-dimethyl.

3 MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

C_6H_{14}

4 THE ALKANE CHECK



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

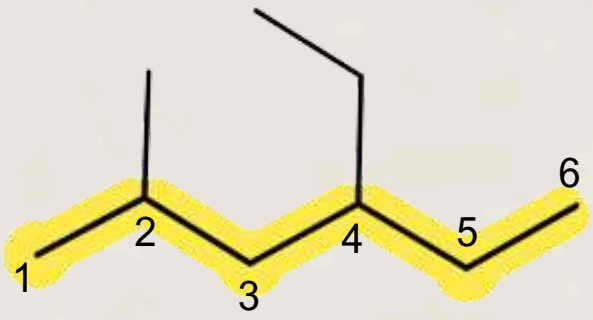
PART A

Structure → Name

A04 - Developing

Counted Formula: C_9H_{20}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 6 C, so hexane. Other 6-C paths exist (through the ethyl), but they lead to the same name.

Numbering: left → right gives 2,4; right → left gives 3,5. 2 < 3, so number from the left.

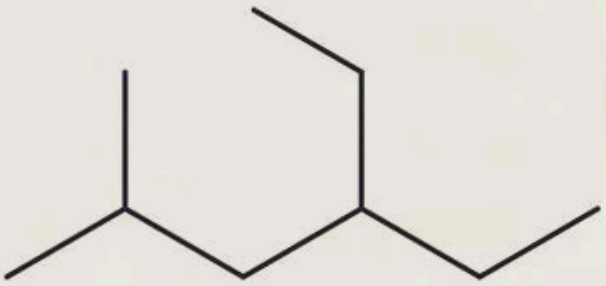
Substituents, alphabetically: ethyl (C4), then methyl (C2).

3 MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

C_9H_{20}

4 THE ALKANE CHECK



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

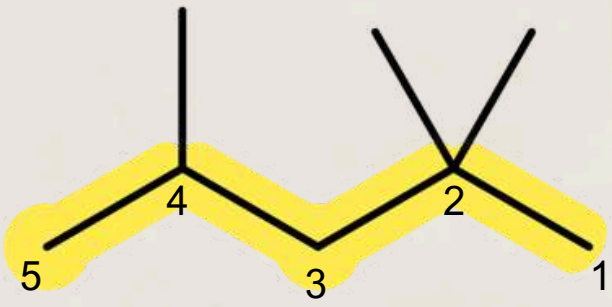
PART A

Structure → Name

A05 - Developing

Counted Formula: C_8H_{18}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 5 C, so pentane.

Numbering: left → right gives 2,4,4; right → left gives 2,2,4. First point of difference is the second term, 2 < 4, so number from the right.

Substituents: three methyls at 2, 2 and 4.

3 MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

C_8H_{18}

4 THE ALKANE CHECK



2,2,4-trimethylpentane

Returned name: 2,2,4-trimethylpentane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

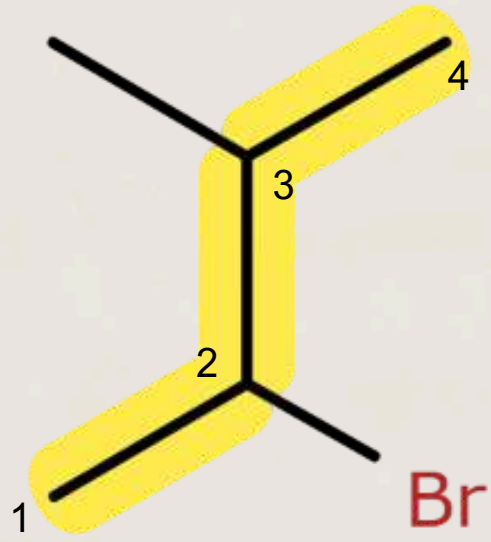
PART A

Structure → Name

A06 - Developing

Counted Formula: $C_5H_{11}Br$

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: 4 C, so butane.

Numbering: from the CH_3 near Br gives Br 2, methyl 3. From the other end gives methyl 2, Br 3. Both sets are {2,3}, so it's a tie. bromo comes before methyl alphabetically, so bromo gets the lower locant.

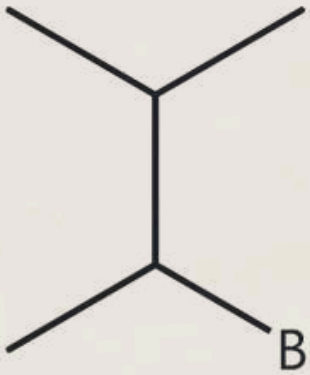
Substituents: bromo (C2), methyl (C3).

3 MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

$C_5H_{11}Br$

4 THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name: 2-bromo-3-methylbutane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

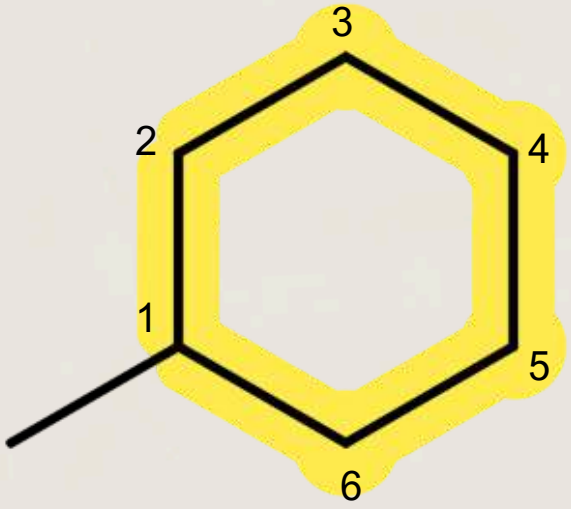
PART A

Structure → Name

A07 – Developing

Counted Formula: C_7H_{14}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: ring of 6 C, so cyclohexane.

Numbering: one substituent only, so it takes C1 and no locant is written.

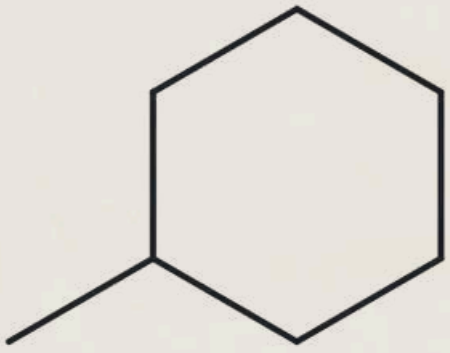
Substituents: methyl.

3 MY NAME (WRITTEN FIRST)

methylcyclohexane

C_7H_{14}

4 THE ALKANE CHECK



methylcyclohexane

Returned name: methylcyclohexane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

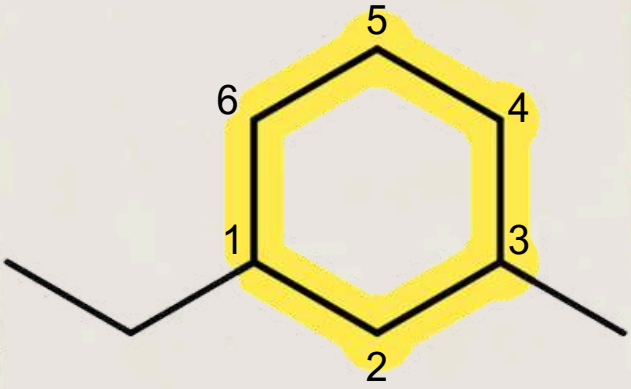
PART A

Structure → Name

A08 - Challenge

Counted Formula: C_9H_{18}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: ring of 6 C, so cyclohexane.

Numbering: the substituents are 1,3 apart whichever way you go round, so the locant sets tie at {1,3}. Ethyl comes before methyl alphabetically, so ethyl gets C1.

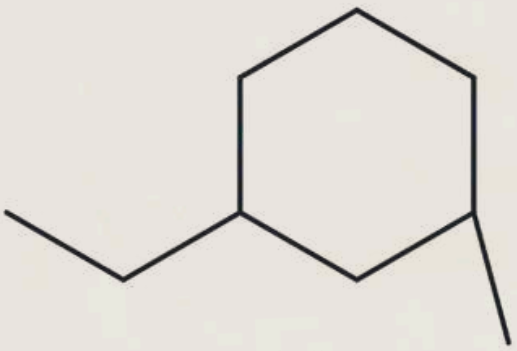
Substituents: ethyl (C1), methyl (C3).

3 MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

C_9H_{18}

4 THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

Returned name: 1-ethyl-3-methylcyclohexane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

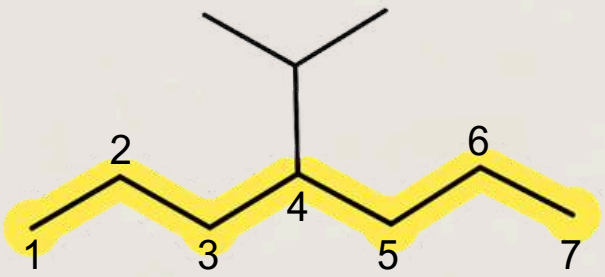
PART A

Structure → Name

A09 - Challenge

Counted Formula: $C_{10}H_{22}$

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain is 7 C, so heptane. The path through the branch is only 6 C, so it loses.

Numbering: the branch is on C4 from either end, so it's a tie.

Substituents: isopropyl (also written propan-2-yl or (1-methylethyl)) at C4. ***Appendix B.3***

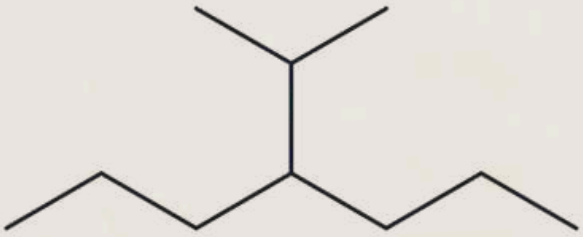
3 MY NAME (WRITTEN FIRST)

4-isopropylheptane

4-(1-methylethyl)heptane

$C_{10}H_{22}$

4 THE ALKANE CHECK



2-methyl-3-propylhexane

Returned name: 2-methyl-3-propylhexane

IT'S NOT A MATCH!

The Alkane website identified the structure as 2-methyl-3-propylhexane, but the longest continuous chain contains 7 carbon atoms, making heptane the correct parent chain. The isopropyl group is attached to carbon 4. Thus, the correct IUPAC name is 4-isopropylheptane.

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

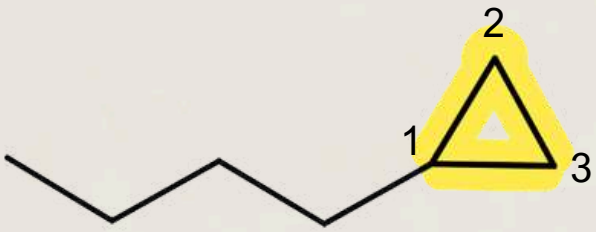
PART A

Structure → Name

A10 - Challenge

Counted Formula: C_7H_{14}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: a ring is present, so the ring is the parent, giving cyclopropane (3 C).

Numbering: one substituent only, so no locant is needed.

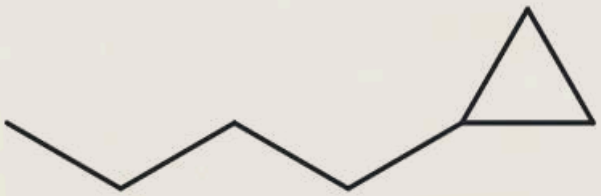
Substituents: butyl (4 C straight chain).

3 MY NAME (WRITTEN FIRST)

butylcyclopropane

C_7H_{14}

4 THE ALKANE CHECK



butylcyclopropane

Returned name: butylcyclopropane

IT'S A MATCH!

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

PART B

Name → Structure

B01 - Foundation

2-methylpropane

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NAME DECODED

Parent: propane → 3 carbon, chain

Substituents: methyl on C2

Expected formula:
3 C + 1 C = 4 C → C_4H_{10}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_4H_{10}

Open Chain: C_nH_{2n+2},
n = 4: C_4H_{10} ✓

4 BACK — CHECK

Returned by The Alkane:

2-methylpropane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B02 - Foundation

3-ethylpentane

1 NAME DECODED

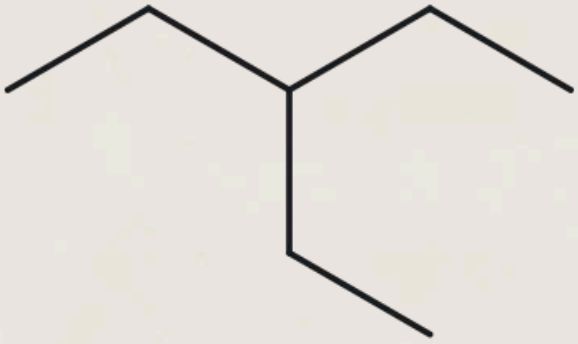
Parent:
pentane → 5 carbon

Substituents:
ethyl on C3

Expected formula:
5 C + 2 C = 7 C
→ C_7H_{16}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_7H_{16}

Open Chain: C_nH_{2n+2},
n = 7: C_7H_{16} ✓

4 BACK — CHECK

Returned by The Alkane:

3-ethylpentane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B03 - Foundation

2,2-dimethylbutane

1 NAME DECODED

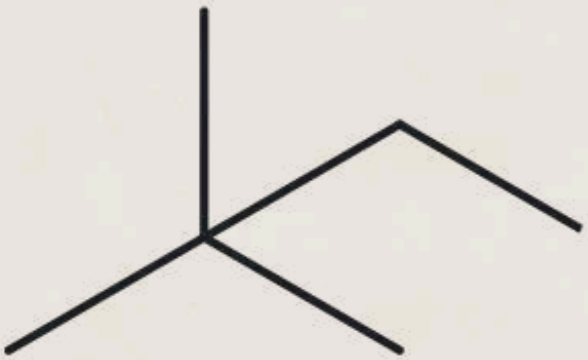
Parent:
butane → 4 carbon

Substituents:
2 × methyl on C2

Expected formula:
4 C + 2 C = 6 C
→ C_6H_{14}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open Chain: C_nH_{2n+2},
n = 6: C_6H_{14} ✓

4 BACK — CHECK

Returned by The Alkane:

2,2-dimethylbutane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B04 - Developing

3-ethyl-2-methylpentane

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NAME DECODED

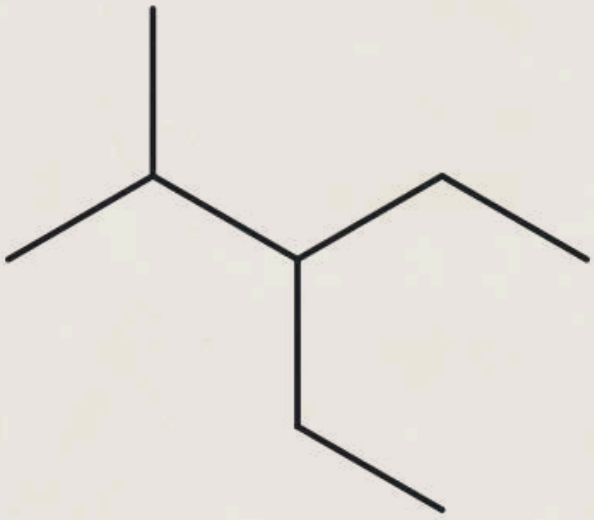
Parent:
pentane → 5 carbon

Substituents: methyl on C2, ethyl on C3

Expected formula:
5 C + 1 C + 2 C = 8 C
→ C_8H_{18}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_8H_{18}

Open Chain: C_nH_{2n+2},
n = 8: C_8H_{18} ✓

4 BACK — CHECK

Returned by The Alkane:

3-ethyl-2-methylpentane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B05 - Developing

1-chloro-2-methylpropane

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NAME DECODED

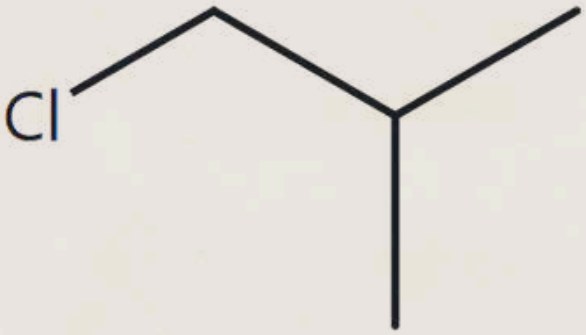
Parent:
propane → 3 carbon

Substituents: chloro on C1, methyl on C2

Expected formula:
3 C + 1 C = 4 C
→ C_4H_{10}, Cl replaces one H → C_4H_9Cl

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_4H_9Cl

Open Chain: C_nH_{2n+2}, n = 4: C_4H_{10}, minus one H for Cl = C_4H_9Cl ✓

4 BACK — CHECK

Returned by The Alkane:

1-chloro-2-methylpropane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B06 - Developing

1,2-dimethylcyclopentane

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NAME DECODED

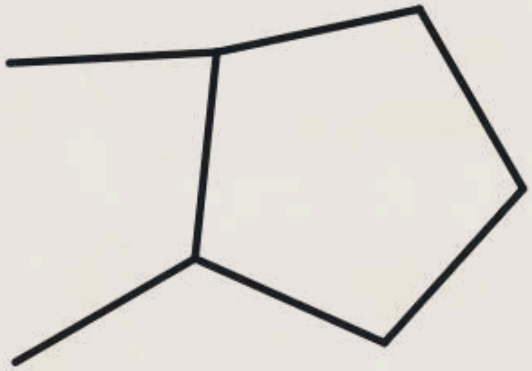
Parent:
cyclopentane → 5 carbon, ring

Substituents: methyl on C1 and C2

Expected formula:
5 C + 2 C = 7 C; one ring → C_7H_{14}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_7H_{14}

One ring: C_nH_{2n},
n = 7: = C_7H_{14} ✓

4 BACK — CHECK

Returned by The Alkane:

1,2-dimethylcyclopentane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART B

Name → Structure

B07 - Developing

4-ethyl-3,3-dimethylheptane

1 NAME DECODED

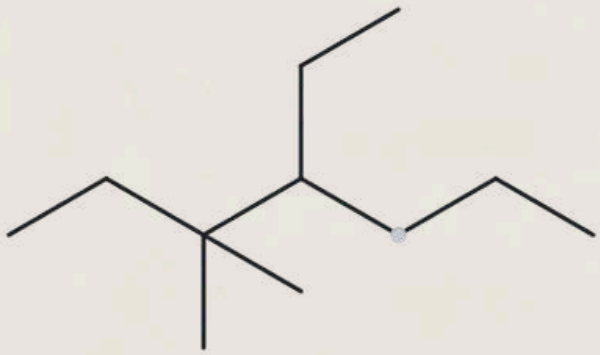
Parent:
heptane → 7 carbon

Substituents: 2 × methyl on C3, ethyl on C4

Expected formula:
7 C + 2 C + 2 C = 11 C
→ $C_{11}H_{24}$

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$

Open Chain: C_nH_{2n+2},
n = 11: = $C_{11}H_{24}$ ✓

4 BACK — CHECK

Returned by The Alkane:

4-ethyl-3,3-dimethylheptane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

PART B

Name → Structure

B08 - Challenge

tert-butylcyclohexane

1 NAME DECODED

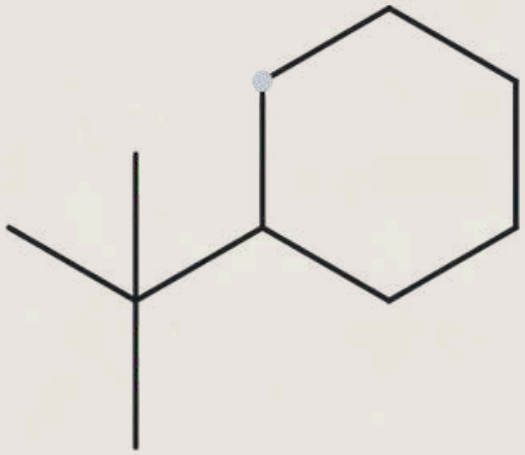
Parent:
cyclohexane → 6 carbon, ring

Substituents: tert-butyl (4 C) on C1

Expected formula:
6 C + 4 C = 10 C; one ring → $C_{10}H_{20}$

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: C_nH_{2n},
n = 10: = $C_{10}H_{20}$ ✓

4 BACK — CHECK

Returned by The Alkane:

(1,1-dimethylethyl)cyclohexane

IT'S A MATCH!

Same compound, different accepted substituent name.

According to Appendix B.3 (Substituent Names and Their Equivalents), tert-butyl is the common name, while the Alkane website uses 1,1-dimethylethyl, which is its systematic equivalent.

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

PART B

Name → Structure

B09 - Challenge

3-ethyl-1,1-dimethylcyclohexane

1 NAME DECODED

Parent:
cyclohexane → 6 carbon, ring

Substituents: 2 × methyl on C1, ethyl on C3

Expected formula:
6 C + 2 C + 2 C = 10 C;
one ring → $C_{10}H_{20}$

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: C_nH_{2n},
n = 10: = $C_{10}H_{20}$ ✓

4 BACK — CHECK

Returned by The Alkane:

3-ethyl-1,1-dimethylcyclohexane

IT'S A MATCH!

Same as the given name!

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded

PART A
PART B
TO BE CONTINUED
TO BE CONTINUED
TO BE CONTINUED

PART B

Name → Structure

BIO - Challenge

2-ethylbutane

PART A

PART B

TO BE CONTINUED

TO BE CONTINUED

TO BE CONTINUED

1 NAME DECODED

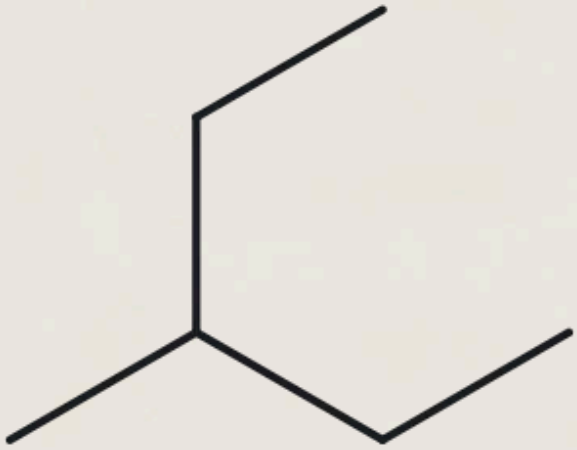
Parent:
butane → 4 carbon

Substituents: ethyl on C2 of butane

Expected formula:
4 + 2 = 6 C → C_6H_{14}

2 BOND LINE STRUCTURE

THE ALKANE



3 FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open Chain: C_nH_{2n+2},
n = 6: C_6H_{14} ✓

4 BACK — CHECK

Returned by The Alkane:

3-methylpentane

IT'S NOT A MATCH!

Not same as the given name!

The ethyl group extends the longest chain to 5 carbons, making pentane the parent chain. The remaining methyl group is attached to C3. Thus, the correct IUPAC name is 3-methylpentane. The mistake was failing to identify the longest continuous chain.

VERIFICATION STRIP

- ☑ parent length match
- ☑ substituents position match
- ☑ formula agrees
- ☑ back - check recorded